Course Catalogue

<table>
<tr><th>Module Code</th><th>Semester</th><th>ECTS</th><th>SWS</th><th>Lecture</th><th>Tutorial</th><th>Lab</th></tr>
<tr><td>MECH-M-DUAL-3-AMR</td><td>3</td><td>5</td><td>3</td><td>3</td><td>0</td><td>0</td></tr>
<tr><td>Course Name</td><td colspan="6">Autonomous Mobile Robotics</td></tr>
<tr><td>Lecturer</td><td colspan="6">D. T. McGuiness, Ph.D (Daniel.McGuiness@mci.edu) (4A-434c)</td></tr>
<tr><td>Study Programme</td><td colspan="6">Mechatronik Automation Robotics AI</td></tr>
<tr><td>Official Name</td><td colspan="3">Autonome mobile Robotik</td><td colspan="2">Course Language</td><td>English</td></tr>
<tr><td>Lecture Prerequisites</td><td colspan="6">The student should have ... After completion, the student will have ...</td></tr>
<tr><td>Course Objectives</td><td colspan="6">The goal of this lecture is ...</td></tr>
<tr><td>Primary Course Content</td><td colspan="6">Lecture Homepage on GitHub | WebBook</td></tr>
<tr><td>Secondary Course Content(s)</td><td colspan="6">A very informal journey through ROS 2 by M. M. Bassa , A Concise Introduction to Robot Programming with ROS2 by F. M. Rico , Programming Principles and Practice using C++ by B. Stroustrup ,</td></tr>
<tr><td>Homework(s) and Project(s)</td><td colspan="6">Personal Assignment A (50) Personal Assignment B (50)</td></tr>
<tr><td rowspan="3">Assessment Criteria</td><td colspan="3">Assignment Type</td><td>Effect</td><td colspan="2">Count</td></tr>
<tr><td colspan="3">Personal Assignment A</td><td>50</td><td colspan="2">1</td></tr>
<tr><td colspan="3">Personal Assignment B</td><td>50</td><td colspan="2">1</td></tr>
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Lecture Structure

Order	Topic	Units	Self Study
1	Locomotion	4	8
	Introduction | Legged Mobile Robots | Wheeled Mobile Robots		
2	Mobile Robot Kinematics	4	8
	Introduction | Kinematic Models and Constraints		
3	Theory of Probability	4	8
	Introduction | Experiments and Outcomes | Probability | Permutations and Combinations | Random Variables and Probability Distributions | Mean and Variance of a Distribution | Binomial, Poisson, and Hypergeometric Distributions | Normal Distribution | Distribution of Several Random Variables		
4	Statistical Methods	4	8
	Introduction | Point Estimation of Parameters | Confidence Intervals | Testing of Hypotheses and Making Decisions | Quality Control | Acceptance Sampling | Goodness of Fit | Non-parametric Tests | Regression and Correlation | Bayesian Statistics		
5	Mobile Robot Localisation	4	8
	Introduction | The problems of Noise and Aliasing | Localisation v. Hard-Coded Navigation | Representing Belief | Representing Maps | Probabilistic Map-Based Localisation | Other Examples of Localisation Methods | Building Maps		
6	Welcome to Linux	2	8
	Learning the Linux Command Line | Installation | Docker		
7	Command Line Fundamentals	4	8
	Introduction | The Structure of Commands | Helpful Keyboard Shortcuts for the Terminal | When you need help with Commands | Additional Information		
8	Working with Files and Folders	2	8
	Introduction | A Detailed Look in ls Command | Creating and Removing Folders | Move, Copy and Delete Files and Folders | Role to Users and sudo | File Permissions | Hard and Symbolic Links | The Linux File System | Common Command-Line Tools and Tasks | Advanced Topics		
9	Installation	1	8
	ROS 2 Humble Hawksbill | Auto-Install Script		
10	ROS Concepts	4	8

Continued on next page

Order	Topic	Units	Self Study
	Introduction | Publisher and Subscriber Architecture | Nodes - The Building Blocks | The Discovery Process | Communication Between Nodes | Topics | Services | Actions | Parameters | Working with Command Line | Launch File | Intermediate Concepts | Configuring Quality of Service | Advanced Concepts		
11	Command Line Tools	4	8
	Setting the Environment | Turtles and Graphs | A Deeper Look into Nodes | Working with Topics | Working with Services | Working with Parameters | A Practical Look into Actions | Launching Nodes		
12	Client Libraries	4	8
	Getting Started with Colcon | Creating a Workspace | Creating a Package | Writing a Simple Publisher and Subscriber | Writing a Simple Service and Client | Creating Custom msg and srv Files | Using Parameters in a Class | Managing Dependencies | Creating an Action | Writing an Action Server and Client | Writing a Launch File		
13	Transform Library	4	8
	A Gentle Introduction | Writing a Static Broadcaster | Writing a Listener | Adding a Frame | Writing a Broadcaster		
14	**Sum**	45	90

- Any major announcements will be made on SAKAI regarding any possible date/content/structural changes for the assignment(s), exam(s).
- Any lecture material will be posted at the lectures corresponding GitHub home-page. The link will be present on the lectures SAKAI homepage.
- If there are any questions regarding course content/exams/assignments please do not refrain from contacting me (Daniel.McGuiness@mci.edu).